

Congestion-Adaptive Betweenness Centrality via Queue-Theoretic Routing Dynamics

Supplementary Material

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I. PROOFS FOR SYSTEM-OPTIMAL CABC

A. Proof of Lemma 1

We first compute the second derivative of $l(\lambda, \mu)$ with respect to λ . Recall that

$$l(\lambda, \mu) = \lambda t(\lambda, \mu) = \frac{\lambda}{\mu} \left(1 + \frac{\lambda(c_a^2 + c_s^2)/2}{\mu - \lambda} \right), \quad 0 < \lambda < \mu. \quad (1)$$

Using the identity $\lambda^2/(\mu - \lambda) = -(\lambda + \mu) + \mu^2/(\mu - \lambda)$, differentiating with respect to λ yields

$$\frac{\partial l}{\partial \lambda}(\lambda, \mu) = \frac{1}{\mu} \left(1 - \frac{c_a^2 + c_s^2}{2} \right) + \frac{(c_a^2 + c_s^2)\mu}{2(\mu - \lambda)^2}, \quad (2)$$

and differentiating again gives

$$\frac{\partial^2 l}{\partial \lambda^2}(\lambda, \mu) = \frac{(c_a^2 + c_s^2)\mu}{(\mu - \lambda)^3}. \quad (3)$$

Since $0 < \lambda < \mu$, the denominator is positive and finite, hence

$$\frac{\partial^2 l}{\partial \lambda^2}(\lambda, \mu) > 0,$$

which establishes convexity.

To determine the strong convexity constant, observe that the mapping $\lambda \mapsto \frac{\mu}{(\mu - \lambda)^3}$ is strictly increasing on $(0, \mu)$. Thus its minimum occurs at $\lambda = 0$, yielding

$$\inf_{0 < \lambda < \mu} \frac{\partial^2 l}{\partial \lambda^2}(\lambda, \mu) = \frac{(c_a^2 + c_s^2)\mu}{\mu^3} = \frac{c_a^2 + c_s^2}{\mu^2}. \quad (4)$$

Therefore $l(\lambda, \mu)$ is strongly convex on $(0, \mu)$ with

$$m_l = \frac{c_a^2 + c_s^2}{\mu^2}. \quad \blacksquare \quad (5)$$

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The simulation code repository can be found at: <https://github.com/rishirani/CABC>

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B. Proof of Theorem 1

Since $f(\lambda, \mu) = \sum_{v \in V} l(\lambda_v, \mu_v)$ is separable, its Hessian in λ is diagonal, with v -th entry $\partial^2 l / \partial \lambda_v^2(\lambda_v, \mu_v)$. By Lemma 1,

$$\frac{\partial^2 l}{\partial \lambda_v^2}(\lambda_v, \mu_v) \geq \frac{c_a^2 + c_s^2}{\mu_v^2} \geq \frac{c_a^2 + c_s^2}{\max_{u \in V} \mu_u^2} = m_f, \quad \forall v \in V, \quad (6)$$

so $f(\lambda, \mu)$ is m_f -strongly convex in λ .

Let $A \in \{0, 1\}^{|V| \times k|E|}$ denote the node-edge incidence matrix so that $\lambda = A\mathbf{x} + \mathbf{s}$. By the chain rule, $\nabla_{\mathbf{x}} f(\mathbf{x}) = A^\top \nabla_{\lambda} f(\lambda)$. Since $AA^\top = \text{diag}(k \deg_{in}(v))$, for any $\mathbf{z} \in \mathbb{R}^{|V|}$,

$$\|A^\top \mathbf{z}\|_2^2 = \mathbf{z}^\top (AA^\top) \mathbf{z} \geq \left(\min_{v \in V} k \deg_{in}(v) \right) \|\mathbf{z}\|_2^2. \quad (7)$$

Taking $\mathbf{z} = \nabla_{\lambda} f(\lambda)$ and using that m_f -strong convexity implies the Polyak–Łojasiewicz inequality $\|\nabla_{\lambda} f(\lambda)\|_2^2 \geq 2m_f(f(\lambda, \mu) - f(\lambda^*, \mu))$,

$$\begin{aligned} \|\nabla_{\mathbf{x}} f(\mathbf{x})\|_2^2 &\geq \left(\min_{v \in V} k \deg_{in}(v) \right) \|\nabla_{\lambda} f(\lambda)\|_2^2 \\ &\geq 2m_f^{PL}(f(\lambda, \mu) - f(\lambda^*, \mu)), \end{aligned} \quad (8)$$

with the Polyak–Łojasiewicz constant being

$$m_f^{PL} = \frac{(c_a^2 + c_s^2) \min_{v \in V} k \deg_{in}(v)}{\max_{v \in V} \mu_v^2} \quad \blacksquare. \quad (9)$$

C. Proof of Lemma 2

Using the same expression as before,

$$\frac{\partial^2 l}{\partial \lambda^2}(\lambda, \mu) = \frac{(c_a^2 + c_s^2)\mu}{(\mu - \lambda)^3}. \quad (10)$$

The function $(\mu - \lambda)^{-3}$ is strictly increasing on $(0, \bar{\rho}\mu)$, so the supremum occurs at $\lambda = \bar{\rho}\mu$, giving

$$\sup_{0 < \lambda < \bar{\rho}\mu} \frac{\partial^2 l}{\partial \lambda^2} = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^3}. \quad (11)$$

Since the Lipschitz smoothness constant equals the supremum of the Hessian,

$$\eta = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^3}. \quad \blacksquare \quad (12)$$

D. Proof of Theorem 2

From Lemma 2,

$$\frac{\partial^2 l}{\partial \lambda_v^2}(\lambda_v, \mu_v) \leq \frac{c_a^2 + c_s^2}{\mu_v^2(1 - \bar{\rho})^3} =: D_v. \quad (13)$$

Let $A \in \{0, 1\}^{|V| \times k|E|}$ denote the node-edge incidence matrix, so $\lambda = A\mathbf{x} + \mathbf{s}$ and the Hessian of the objective is

$$\nabla_{\mathbf{x}}^2 f(\mathbf{x}) = A^\top \text{diag}(D_v)A.$$

Since every column of A has a single nonzero entry, grouping columns by target node makes $A^\top \text{diag}(D_v)A$ block-diagonal, with the block for node v equal to $D_v \mathbf{1}_{n_v} \mathbf{1}_{n_v}^\top$, where $n_v = k \deg_{in}(v)$ is the number of commodity-edge columns terminating at v . This rank-one block has a single nonzero eigenvalue $D_v n_v$ (eigenvector $\mathbf{1}_{n_v}$), so

$$\lambda_{\max}(\nabla_{\mathbf{x}}^2 f) = \max_{v \in V} D_v n_v = k(c_a^2 + c_s^2) \max_{v \in V} \frac{\deg_{in}(v)}{\mu_v^2(1 - \bar{\rho})^3}. \quad (14)$$

Thus f is Lipschitz smooth with

$$\eta_f = k(c_a^2 + c_s^2) \max_{v \in V} \frac{\deg_{in}(v)}{\mu_v^2(1 - \bar{\rho})^3}. \quad \blacksquare$$

E. Proof of Theorem 3

By Theorem 2, f is η_f -smooth in $\mathbf{x}$, and by Theorem 1, f satisfies the Polyak–Łojasiewicz inequality in $\mathbf{x}$ with constant m_f^{PL} . Projected gradient descent on an η_f -smooth function satisfying this inequality achieves geometric convergence in objective value [1], [2],

$$f(\lambda_n, \mu) - f(\lambda^*, \mu) \leq \left(1 - \frac{1}{\kappa_f}\right)^n (f(\lambda_0, \mu) - f(\lambda^*, \mu)), \quad (15)$$

where $\kappa_f = \frac{\eta_f}{m_f^{PL}}$. Since $f(\lambda, \mu)$ is m_f -strongly convex in λ (Theorem 1),

$$\begin{aligned} \|\lambda_n - \lambda^*\|_2^2 &\leq \frac{2}{m_f} (f(\lambda_n, \mu) - f(\lambda^*, \mu)) \\ &\leq \frac{2}{m_f} \left(1 - \frac{1}{\kappa_f}\right)^n (f(\lambda_0, \mu) - f(\lambda^*, \mu)). \quad \blacksquare \end{aligned} \quad (16)$$

II. PROOFS FOR USER EQUILIBRIUM CABG

A. Proof of Lemma 3

We compute the second derivative of $T(\lambda, \mu)$ with respect to λ . Recall that the integral of Kingman's approximation for sojourn time is

$$T(\lambda, \mu) = \int_0^\lambda t(s, \mu) ds, \quad (17)$$

$$\text{where, } t(\lambda, \mu) = \frac{1}{\mu - \lambda} \left(1 + \frac{c_a^2 + c_s^2}{2}\right) - \frac{c_a^2 + c_s^2}{2\mu}. \quad (18)$$

Differentiating under the integral gives

$$\frac{\partial T}{\partial \lambda}(\lambda, \mu) = t(\lambda, \mu), \quad (19)$$

and differentiating again yields

$$\frac{\partial^2 T}{\partial \lambda^2}(\lambda, \mu) = \frac{\partial t}{\partial \lambda}(\lambda, \mu) = \frac{c_a^2 + c_s^2}{(\mu - \lambda)^2}. \quad (20)$$

Because $\lambda < \mu$, the denominator $(\mu - \lambda)^2$ is positive. Thus

$$\frac{\partial^2 T}{\partial \lambda^2}(\lambda, \mu) > 0,$$

proving convexity.

To obtain the strong convexity constant, note that $(\mu - \lambda)^{-2}$ is strictly increasing on $(0, \mu)$, achieving its minimum at $\lambda = 0$. Thus

$$\min_{0 < \lambda < \mu} \frac{\partial^2 T}{\partial \lambda^2} = \frac{c_a^2 + c_s^2}{\mu^2}. \quad (21)$$

Hence $T(\lambda, \mu)$ is strongly convex with constant

$$m = \frac{c_a^2 + c_s^2}{\mu^2}. \quad \blacksquare \quad (22)$$

B. Proof of Theorem 4

Since $g(\lambda, \mu) = \sum_{v \in V} T(\lambda_v, \mu_v)$ is separable, its Hessian in λ is diagonal, with v -th entry $\partial^2 T / \partial \lambda_v^2(\lambda_v, \mu_v)$. By Lemma 3,

$$\frac{\partial^2 T}{\partial \lambda_v^2}(\lambda_v, \mu_v) \geq \frac{c_a^2 + c_s^2}{\mu_v^2} \geq \frac{c_a^2 + c_s^2}{\max_{u \in V} \mu_u^2} = m_g, \quad \forall v \in V, \quad (23)$$

so $g(\lambda, \mu)$ is m_g -strongly convex in λ .

Let $A \in \{0, 1\}^{|V| \times k|E|}$ denote the node-edge incidence matrix so that $\lambda = A\mathbf{x} + \mathbf{s}$. By the chain rule, $\nabla_{\mathbf{x}} g(\mathbf{x}) = A^\top \nabla_{\lambda} g(\lambda)$. Since $AA^\top = \text{diag}(k \deg_{in}(v))$, for any $\mathbf{z} \in \mathbb{R}^{|V|}$,

$$\|A^\top \mathbf{z}\|_2^2 = \mathbf{z}^\top (AA^\top) \mathbf{z} \geq \left(\min_{v \in V} k \deg_{in}(v)\right) \|\mathbf{z}\|_2^2. \quad (24)$$

Taking $\mathbf{z} = \nabla_{\lambda} g(\lambda)$ and using that m_g -strong convexity implies the Polyak–Łojasiewicz inequality $\|\nabla_{\lambda} g(\lambda)\|_2^2 \geq 2m_g(g(\lambda, \mu) - g(\lambda^*, \mu))$,

$$\|\nabla_{\mathbf{x}} g(\mathbf{x})\|_2^2 \geq \left(\min_{v \in V} k \deg_{in}(v)\right) \|\nabla_{\lambda} g(\lambda)\|_2^2 \quad (25)$$

$$\geq 2m_g^{PL} (g(\lambda, \mu) - g(\lambda^*, \mu)), \quad (26)$$

with the Polyak–Łojasiewicz constant being

$$m_g^{PL} = \frac{(c_a^2 + c_s^2) \min_{v \in V} k \deg_{in}(v)}{\max_{v \in V} \mu_v^2}. \quad \blacksquare \quad (27)$$

C. Proof of Lemma 4

From the derivation above,

$$\frac{\partial^2 T}{\partial \lambda^2}(\lambda, \mu) = \frac{c_a^2 + c_s^2}{(\mu - \lambda)^2}. \quad (28)$$

For $\lambda \in (0, \bar{\rho}\mu)$, the expression is maximized at $\lambda = \bar{\rho}\mu$, giving

$$\sup_{0 < \lambda < \bar{\rho}\mu} \frac{\partial^2 T}{\partial \lambda^2} = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^2}. \quad (29)$$

Since the Lipschitz smoothness constant equals the supremum of the Hessian,

$$\eta = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^2}. \quad \blacksquare \quad (30)$$

D. Proof of Theorem 5

From Lemma 4,

$$\frac{\partial^2 T}{\partial \lambda_v^2}(\lambda_v, \mu_v) \leq \frac{c_a^2 + c_s^2}{\mu_v^2(1 - \bar{\rho})^2} =: D_v. \quad (31)$$

Let $A \in \{0, 1\}^{|V| \times k|E|}$ denote the node-edge incidence matrix, so $\mathbf{\lambda} = A\mathbf{x} + \mathbf{s}$ and the Hessian of the objective is

$$\nabla_{\mathbf{x}}^2 g(\mathbf{x}) = A^\top \text{diag}(D_v) A.$$

Since every column of A has a single nonzero entry, grouping columns by target node makes $A^\top \text{diag}(D_v) A$ block-diagonal, with the block for node v equal to $D_v \mathbf{1}_{n_v} \mathbf{1}_{n_v}^\top$, where $n_v = k \deg_{in}(v)$ is the number of commodity-edge columns terminating at v . This rank-one block has a single nonzero eigenvalue $D_v n_v$ (eigenvector $\mathbf{1}_{n_v}$), so

$$\lambda_{\max}(\nabla_{\mathbf{x}}^2 g) = \max_{v \in V} D_v n_v = k(c_a^2 + c_s^2) \max_{v \in V} \frac{\deg_{in}(v)}{\mu_v^2(1 - \bar{\rho})^2}. \quad (32)$$

Thus g is Lipschitz smooth with

$$\eta_g = k(c_a^2 + c_s^2) \max_{v \in V} \frac{\deg_{in}(v)}{\mu_v^2(1 - \bar{\rho})^2}. \quad \blacksquare$$

E. Proof of Theorem 6

By Theorem 5, g is η_g -smooth in $\mathbf{x}$, and by Theorem 4, g satisfies the Polyak–Łojasiewicz inequality in $\mathbf{x}$ with constant m_g^{PL} . Projected gradient descent on an η_g -smooth function satisfying this inequality achieves geometric convergence in objective value [1], [2], with condition number $\kappa_g = \eta_g/m_g^{PL}$:

$$g(\mathbf{\lambda}_n, \boldsymbol{\mu}) - g(\mathbf{\lambda}^*, \boldsymbol{\mu}) \leq \left(1 - \frac{1}{\kappa_g}\right)^n (g(\mathbf{\lambda}_0, \boldsymbol{\mu}) - g(\mathbf{\lambda}^*, \boldsymbol{\mu})). \quad (33)$$

Since $g(\mathbf{\lambda}, \boldsymbol{\mu})$ is m_g -strongly convex in $\mathbf{\lambda}$ (Theorem 4),

$$\begin{aligned} \|\mathbf{\lambda}_n - \mathbf{\lambda}^*\|_2^2 &\leq \frac{2}{m_g} (g(\mathbf{\lambda}_n, \boldsymbol{\mu}) - g(\mathbf{\lambda}^*, \boldsymbol{\mu})) \\ &\leq \frac{2}{m_g} \left(1 - \frac{1}{\kappa_g}\right)^n (g(\mathbf{\lambda}_0, \boldsymbol{\mu}) - g(\mathbf{\lambda}^*, \boldsymbol{\mu})). \quad \blacksquare \end{aligned} \quad (34)$$

(35)

III. PROOFS FOR OPERATING REGIME ANALYSIS

A. Proof of Theorem 7

Consider

$$l(\lambda, \mu) = \lambda t(\lambda, \mu) = \frac{\lambda}{\mu} \left(1 + \frac{\lambda(c_a^2 + c_s^2)/2}{\mu - \lambda}\right), \quad \lambda < \mu. \quad (36)$$

For fixed μ , expand the denominator in the low-traffic regime $\lambda \ll \mu$ using the geometric series

$$\frac{1}{\mu - \lambda} = \frac{1}{\mu} \frac{1}{1 - \lambda/\mu} = \frac{1}{\mu} \sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^k. \quad (37)$$

Then

$$\frac{\lambda(c_a^2 + c_s^2)/2}{\mu - \lambda} = \frac{\lambda(c_a^2 + c_s^2)}{2\mu} \sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^k \quad (38)$$

$$= \frac{\lambda(c_a^2 + c_s^2)}{2\mu} + \frac{\lambda^2(c_a^2 + c_s^2)}{2\mu^2} + O\left(\frac{\lambda^3}{\mu^3}\right). \quad (39)$$

Multiplying by λ/μ gives

$$l(\lambda, \mu) = \frac{\lambda}{\mu} + O\left(\frac{\lambda^2}{\mu^2}\right). \quad (40)$$

Thus, as $\mu \rightarrow \infty$ (or equivalently $\lambda/\mu \rightarrow 0$),

$$l(\lambda, \mu) = \frac{\lambda}{\mu} + O\left(\frac{1}{\mu^2}\right),$$

which establishes convergence to a linear function in the low-traffic regime. $\blacksquare$

B. Proof of Theorem 8

The integral of Kingman's approximation of sojourn time is

$$\begin{aligned} T(\lambda, \mu) &= \int_0^\lambda t(s, \mu) ds \\ &= \frac{c_a^2 + c_s^2}{2} \ln\left(\frac{\mu}{\mu - \lambda}\right) + \frac{\lambda}{\mu} \left(1 - \frac{c_a^2 + c_s^2}{2}\right), \quad \lambda < \mu. \end{aligned} \quad (41)$$

Using the Taylor expansion for $\ln(1+x)$,

$$\ln\left(\frac{\mu}{\mu - \lambda}\right) = \ln\left(1 + \frac{\lambda}{\mu - \lambda}\right) = \frac{\lambda}{\mu} + \frac{\lambda^2}{2\mu^2} + O\left(\frac{\lambda^3}{\mu^3}\right),$$

we substitute into $T(\lambda, \mu)$ to obtain

$$\begin{aligned} T(\lambda, \mu) &= \frac{\lambda}{\mu} \left(1 - \frac{c_a^2 + c_s^2}{2}\right) + \frac{c_a^2 + c_s^2}{2} \left(\frac{\lambda}{\mu} + \frac{\lambda^2}{2\mu^2} + O\left(\frac{\lambda^3}{\mu^3}\right)\right) \\ &= \frac{\lambda}{\mu} + O\left(\frac{\lambda^2}{\mu^2}\right). \end{aligned} \quad (42)$$

Thus,

$$T(\lambda, \mu) = \frac{\lambda}{\mu} + O\left(\frac{1}{\mu^2}\right),$$

which proves linear behavior in the low-traffic regime. $\blacksquare$

C. Proof of Theorem 9

From Lemma 1, $l(\lambda, \mu)$ is m -strongly convex in λ over $(0, \mu)$ with

$$m = \frac{c_a^2 + c_s^2}{\mu^2}. \quad (44)$$

Applying the strong convexity inequality at $\lambda_0 = 0$,

$$l(\lambda, \mu) \geq l(0, \mu) + \frac{\partial l}{\partial \lambda}(0, \mu) \lambda + \frac{m}{2} \lambda^2, \quad (45)$$

and using $l(0, \mu) = 0$ and $\frac{\partial l}{\partial \lambda}(0, \mu) = \frac{1}{\mu}$ yields the lower bound

$$l(\lambda, \mu) \geq \frac{\lambda}{\mu} + \frac{(c_a^2 + c_s^2)}{2\mu^2} \lambda^2, \quad 0 \leq \lambda < \mu. \quad (46)$$

For the upper bound, restrict to $\lambda \leq \bar{\rho} \cdot \mu$. Lemma 2 shows that l is η -smooth on this interval with

$$\eta = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^3}. \quad (47)$$

The smoothness inequality at $\lambda_0 = 0$ gives

$$l(\lambda, \mu) \leq \frac{\lambda}{\mu} + \frac{\eta}{2} \lambda^2, \quad 0 \leq \lambda \leq \bar{\rho} \mu, \quad (48)$$

which becomes

$$l(\lambda, \mu) \leq \frac{\lambda}{\mu} + \frac{(c_a^2 + c_s^2)}{2\mu^2(1 - \bar{\rho})^3} \lambda^2. \quad (49)$$

Since $\partial l / \partial \lambda > 0$ for $\lambda < \mu$, $l(\lambda, \mu)$ is monotone increasing. Combining the two inequalities yields the stated quadratic upper and lower bounds. ■

D. Proof of Theorem 10

Lemma 3 shows that $T(\lambda, \mu)$ is m -strongly convex with

$$m = \frac{c_a^2 + c_s^2}{\mu^2}. \quad (50)$$

Applying the strong convexity inequality at $\lambda_0 = 0$ gives

$$T(\lambda, \mu) \geq \frac{\lambda}{\mu} + \frac{(c_a^2 + c_s^2)}{2\mu^2} \lambda^2, \quad 0 \leq \lambda < \mu. \quad (51)$$

For the upper bound, restrict again to $\lambda \leq \bar{\rho} \mu$. Lemma 4 yields smoothness constant

$$\eta = \frac{c_a^2 + c_s^2}{\mu^2(1 - \bar{\rho})^2}. \quad (52)$$

The smoothness inequality at $\lambda_0 = 0$ then gives

$$T(\lambda, \mu) \leq \frac{\lambda}{\mu} + \frac{\eta}{2} \lambda^2 = \frac{\lambda}{\mu} + \frac{(c_a^2 + c_s^2)}{2\mu^2(1 - \bar{\rho})^2} \lambda^2. \quad (53)$$

Since $t(\lambda, \mu) > 0$ for $\lambda < \mu$, $T(\lambda, \mu)$ is monotone increasing. Combining the upper and lower bounds produces the stated quadratic bounds. ■

IV. SYNTHETIC LOGISTICAL NETWORK

A. Network Topology of the Logistics Network Model

To complement the Abilene-Internet case study, we evaluate the proposed congestion-adaptive betweenness centrality framework on a geographically structured logistics network. The model represents a regional supply chain with directed commodity flow, hierarchical aggregation, and explicit separation between supply and demand. As in many large-scale engineered network, including communication networks, logistical systems exhibit pronounced structural and dynamical asymmetries, with commodities originating at a limited number of upstream production facilities and flowing through intermediate distribution layers to satisfy geographically distributed demand. This topology therefore provides a realistic abstraction for studying congestion-aware node importance in transportation and supply-chain settings.

The network is organized as a four-tier hierarchy comprising manufacturers, regional distributors (RDs), local distributors (LDs), and retailers, and is partitioned into three geographic regions (West, Central, and East). Within each region, manufacturers connect to all regional distributors, reflecting centralized production feeding multiple aggregation hubs. Regional

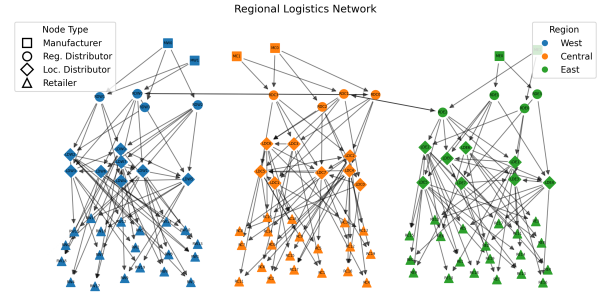


Fig. 1. Four-tier geographically structured logistical network used in the simulation study. Node shapes correspond to supply-chain tier (manufacturer, regional distributor, local distributor, retailer), and node colors denote geographic region (West, Central, East). Bidirectional RD-RD gateway links provide limited cross-regional coordination.

distributors coordinate both downstream distribution and inter-regional transfers, while local distributors handle intra-regional redistribution and retailers act as terminal demand nodes. Limited cross-regional connectivity is introduced exclusively at the RD layer via bidirectional gateway links, enabling inter-regional coordination and transport while preserving predominantly regional flow structure. The logistical topology used in simulation is generated stochastically, with randomized but structured connectivity within and across tiers subject to fixed degree and hierarchy constraints. This controlled randomness avoids artificial symmetry while preserving realistic supply-chain structure, ensuring that node importance is not predetermined by construction but instead emerges from the interaction between topology, geographic partitioning, and congestion dynamics. The resulting directed, capacity-constrained network is illustrated in Fig. 1, where node shapes denote supply-chain tier and node colors indicate geographic region.

External Dynamics and Network Parametrization

We next describe the external supply-demand dynamics and capacity parametrization used to generate congestion in the logistics network. In this setting, network flow represents the transport of physical commodities and is measured in units per week. Manufacturer nodes act as exogenous sources of supply, retailer nodes act as strict sinks with fixed demand requirements, and all intermediate distributor nodes are flow-conserving.

Supply and capacity are intentionally heterogeneous across regions. In particular, manufacturers located in the central region provide 7500 units/week of supply each and are provisioned with capacity 15000 units/week, while manufacturers in the west and east provide 3750 units/week of supply each with capacity 5000 units/week. This asymmetry reflects realistic supply-chain conditions in which production capacity often concentrates in regions that do not coincide with dominant demand centers. Distributor handling capacities are set to 7500 units/week for each regional distributor and 3000 units/week for each local distributor, establishing tier-dependent bottlenecks that couple congestion to the hierarchical organization of the network and induce nontrivial inter-regional transfers through the RD gateway layer. **Because every**

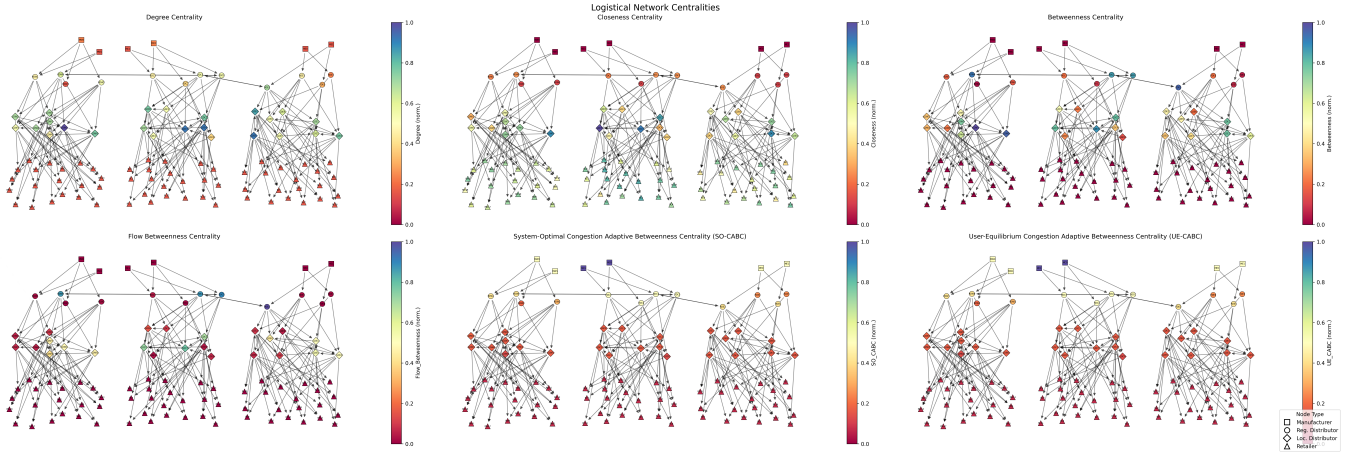


Fig. 2. Comparison of structural centralities (degree, closeness, betweenness, flow-betweenness) with the proposed SO-CABC and UE-CABC measures on the four-tier regional logistics network. The figure spans both columns to accommodate the full set of centrality heatmaps.

regional distributor is assigned the same handling capacity (service rate) $\mu_v = 7500$ units/week irrespective of region or direction, the bidirectional RD–RD gateway links are symmetric in service rate: neither direction of a gateway link is favored a priori by node-level congestion, so any directional asymmetry in the resulting UE-CABC routing arises solely from the demand and topological asymmetries described above (e.g., unequal regional supply/demand shares), rather than from asymmetric queueing at the gateway endpoints. Each retailer demands 500 units/week, and retailer intake capacity is set to 750 units/week to ensure feasibility under moderate congestion.

Congestion-induced delay is modeled using the same queueing-inspired convex proxy introduced in Section 3.2. For the logistics setting, the coefficients of variation are set to $c_a = 2$ for arrivals and $c_s = 0.5$ for service, reflecting smoother demand profiles than Internet traffic while accounting for variability due to transportation delays, scheduling uncertainty, and operational disruptions. Together with the heterogeneous supply, demand, and capacity constraints, this parametrization produces spatially nonuniform congestion patterns that cannot be inferred from topology alone.

Performance Evaluation of Centrality Measures

TABLE I
COMPUTATION TIMES FOR THE CENTRALITY MEASURES ON THE LOGISTICS NETWORK.

Centrality Measure	Runtime (s)
Degree	0.000184
Closeness	0.001505
Betweenness	0.001811
Flow-Betweenness	12.926730
SO-CABC	0.012921
UE-CABC	0.008705

We now compare classical structural centrality measures with the proposed congestion-adaptive betweenness centralities on the logistics network. The resulting node-importance maps are shown in Fig. 2, and the corresponding computation times are reported in Table I.

Degree Centrality: Degree centrality assigns the highest importance to local distributors and regional distributors, while manufacturers receive uniformly low scores. This behavior is driven purely by local fan-in and fan-out and fails to reflect the upstream role of manufacturers as the sole sources of commodity supply. As a result, degree centrality conflates aggregation with operational importance and does not recover the hierarchical structure of the supply chain. Computationally, degree centrality is the fastest baseline, requiring only 1.84×10^{-4} seconds, but this negligible cost comes at the expense of dynamical fidelity.

Closeness Centrality: Closeness centrality similarly favors local distributors, but in a non-uniform and erratic manner. Although this metric partially distinguishes tiers, it does not produce a clear hierarchical ordering across manufacturers, distributors, and retailers. Nodes with similar hop-distance profiles but fundamentally different supply-chain roles receive comparable importance, indicating that distance-based metrics are insensitive to the directed and asymmetric nature of commodity flow. Its runtime remains very small at 1.505×10^{-3} seconds, reflecting the simplicity of shortest-path-based structural analysis.

Betweenness Centrality: Betweenness centrality partially captures coordination effects by highlighting regional distributor gateway nodes that mediate inter-regional transport. However, these gateways are not consistently identified as the most important nodes. Several local distributors outrank gateway regional distributors due to their frequent appearance on shortest paths between regional retailers, even though their operational impact is limited to intra-regional redistribution. This illustrates a fundamental limitation of shortest-path-based centrality: it emphasizes geometric bridging rather than the ability to reconcile global supply and demand under capacity constraints. The runtime is still modest at 1.811×10^{-3} seconds, but the additional cost over degree and closeness does not translate into an importance profile that reflects the operational hierarchy of the logistics system.

Flow-Betweenness Centrality: Flow-betweenness centrality provides a stronger baseline than shortest-path be-

tweenness in this logistics setting. In particular, it is less erratic in its importance scaling and more consistently ranks regional distributors above local distributors. This improvement arises because flow betweenness accounts for capacity-feasible transfer: regional distributors can support substantially larger feasible flows than local distributors due to their higher handling capacities, and are therefore assigned greater importance. However, it still does not recover the intended hierarchical ordering of the supply chain. Although it incorporates capacity constraints, it is not fully aligned with the directional source-to-sink commodity structure of the logistics problem, in which flow moves from manufacturers to retailers through the intermediate distribution hierarchy. As a result, flow betweenness improves upon shortest-path betweenness, but it still does not produce the clear monotonic layered importance exhibited by SO-CABC and UE-CABC. This limitation is accompanied by a substantial computational burden: flow betweenness requires 12.926730 seconds, several orders of magnitude slower than all other measures considered here, because it effectively requires solving a combinatorial number of pairwise flow programs, scaling at the level of $\mathcal{O}(N^2)$ source–sink evaluations.

Congestion Adaptive Betweenness Centralities: In contrast, both system-optimal and user-equilibrium congestion-adaptive betweenness centralities recover the hierarchical structure of the logistics network, with importance increasing monotonically from retailers to local distributors, regional distributors, and manufacturers. Regional distributors—particularly gateway nodes—are consistently identified as the most important coordination points, while manufacturers in the central region are assigned higher importance than their counterparts in the west and east due to their greater supply contribution. Notably, gateway regional distributors are important but not dominant, reflecting the fact that each region remains largely self-sufficient: the west and east regions generate approximately 25% of total supply while accounting for 33% of demand, requiring only modest inter-regional transfers. From a computational standpoint, SO-CABC and UE-CABC require only 0.012921 and 0.008705 seconds, respectively, despite modeling all relevant flows concurrently and explicitly accounting for congestion. Unlike flow betweenness, they avoid solving a separate program for every source–sink pair by expanding the optimization state space to represent the relevant commodity flows jointly in a single convex program.

Overall, these results demonstrate that congestion-adaptive betweenness centrality identifies nodes whose importance arises from supply–demand asymmetry, capacity limitations, and congestion dynamics, rather than purely from structural position. The qualitative agreement between SO-CABC and UE-CABC further suggests that, in this logistics setting, routing behavior is dominated by physical constraints rather than strategic path selection. At the same time, Table I shows that this improved modeling fidelity is achieved with runtimes that remain dramatically smaller than flow betweenness, indicating that the proposed formulations provide both a more appropriate notion of operational importance and a more scalable computational framework.

V. SENSITIVITY ANALYSIS ON THE 7×7 GRID

In this section, we provide additional experiments on the 7×7 grid benchmark of Section 6.2 in order to examine the sensitivity of the induced centrality profiles to the congestion parameters of the model. The purpose of these experiments is not to demonstrate abrupt instability of the proposed rankings under small perturbations, but rather to verify that the profiles vary in a structurally interpretable manner as the queueing penalty and utilization constraint are changed.

The experimental setup follows the grid construction of the main text, with the source located at the center node of the left-most column and the sink located at the center node of the right-most column. To suppress spurious boundary bottlenecks and isolate the effect of interior rerouting, all nodes in the left-most and right-most columns are assigned capacity 1000, while all interior nodes are assigned capacity 100. In all cases, the utilization constraint is enforced at every node, including the source and sink.

We consider two complementary parameter sweeps. In the first, we vary the variability weight

$$K \triangleq \frac{c_a^2 + c_s^2}{2}, \quad (54)$$

while fixing the utilization cap $\bar{\rho}$. In the second, we vary $\bar{\rho}$ while fixing K . These parameters play distinct roles in the model. The parameter K controls the strength of the queueing-induced nonlinear penalty in the routing objective and therefore acts as a soft congestion-sensitivity parameter. By contrast, $\bar{\rho}$ imposes a direct upper bound on admissible node utilization and therefore acts as a hard feasibility constraint.

A. Sensitivity with Respect to K

We first fix the demand at $d = 10$ and the utilization cap at $\bar{\rho} = 0.99$, and vary the variability weight over the set

$$K \in \{0, 1, 2, 3\}. \quad (55)$$

The resulting profiles are shown in Fig. 3, where the top row corresponds to SO-CABC and the bottom row corresponds to UE-CABC.

At $K = 0$, both formulations reduce to the linear-routing limit and place essentially all centrality mass on the middle horizontal geodesic joining source and sink. This agrees with the low-traffic behavior established in Section 5 and observed in Section 6.2: when the nonlinear queueing contribution is absent, the induced profile collapses to the shortest central corridor. As K increases, both SO-CABC and UE-CABC broaden away from this corridor and assign non-negligible importance to adjacent rows. This effect is monotone and spatially coherent across the sweep, indicating that the parameter K does not introduce erratic changes in the profile, but instead gradually promotes route diversification.

A second feature of Fig. 3 is the persistent separation between the two routing regimes. For each fixed value of K , the SO-CABC profile spreads more strongly across the grid than the corresponding UE-CABC profile. In particular, the rows immediately above and below the center acquire substantial

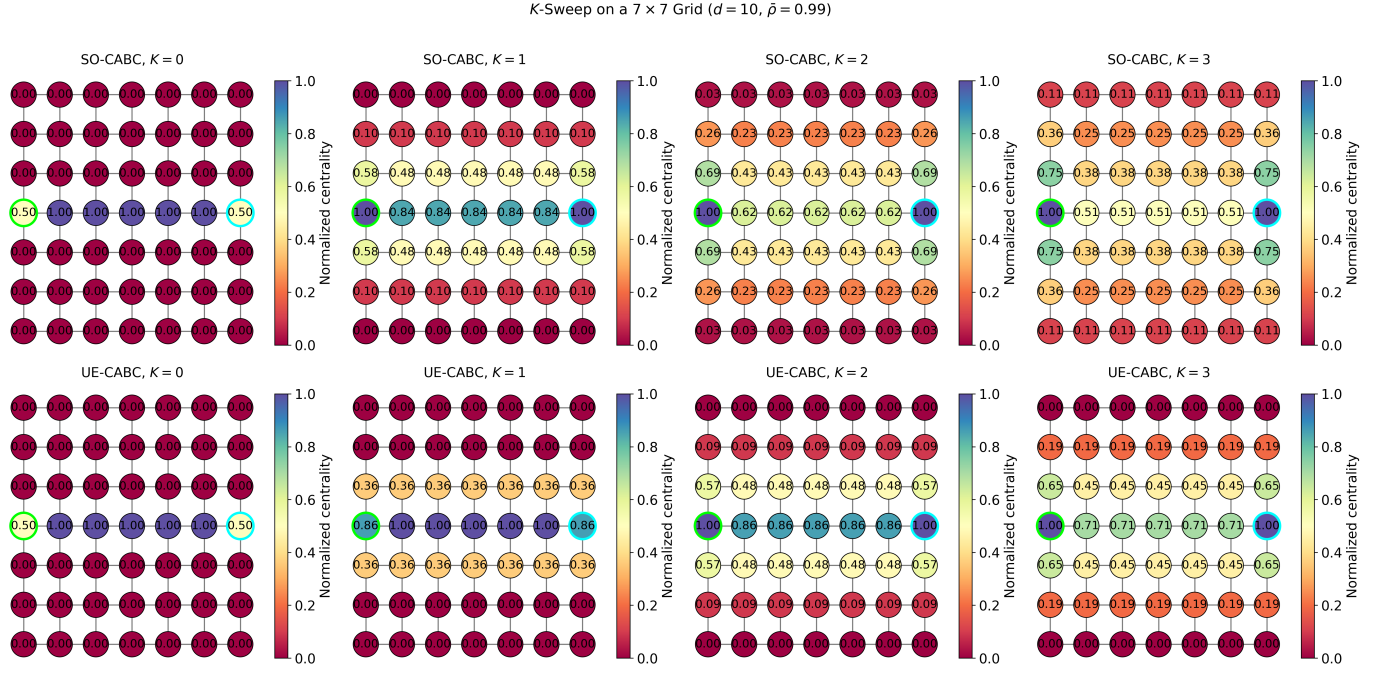


Fig. 3. Sensitivity of the induced centrality profiles to the variability weight K on the 7×7 grid. The demand is fixed at $d = 10$ and the utilization cap is fixed at $\bar{\rho} = 0.99$. The top row shows SO-CABC and the bottom row shows UE-CABC. Increasing K produces a gradual broadening of the centrality profile, with a stronger spreading effect under the system-optimal formulation.

weight under the system-optimal formulation at smaller values of K than under the user-equilibrium formulation. This is consistent with the interpretation developed in the main text: the SO objective internalizes congestion globally and therefore penalizes concentrated loading more aggressively, whereas the UE objective reflects only the individual marginal delay seen by infinitesimal users and thus preserves a stronger preference for the shortest central route.

Taken together, these observations support the interpretation of K as a soft load-balancing parameter. Increasing K strengthens the queueing-induced curvature of the routing objective and thereby broadens the induced centrality profile in a gradual and interpretable fashion. The effect is present under both routing regimes, but is more pronounced under SO-CABC than under UE-CABC.

To relate this sweep to baseline structural betweenness, note that on the present grid the classical shortest-path betweenness profile is concentrated on the central horizontal corridor connecting source and sink. The $K = 0$ limit is consistent with this geodesic-dominated structure. As K increases, the induced SO-CABC and UE-CABC profiles broaden onto adjacent near-geodesic routes, so the resulting node ranking progressively departs from the structural betweenness ranking. This difference should be interpreted carefully: the departure from structural betweenness is driven not only by the congestion-strength parameter K , but also by the fact that CABC is defined with explicitly prescribed source-sink commodities rather than an implicit all-pairs traffic model. Accordingly, the sensitivity sweep isolates only the first effect, namely how strongly congestion reshapes the ranking once the operational traffic pattern has been fixed.

B. Sensitivity with Respect to $\bar{\rho}$

We next fix the demand at $d = 100$ and the variability weight at $K = 0$, and vary the utilization cap over the set

$$\bar{\rho} \in \{1, 0.6, 0.3, 0.15\}. \quad (56)$$

The resulting profiles are shown in Fig. 4, again with SO-CABC in the top row and UE-CABC in the bottom row.

When $\bar{\rho} = 1$, the utilization cap is effectively inactive, and both formulations recover the narrow middle-row profile associated with unconstrained linear routing. As $\bar{\rho}$ is tightened, the admissible loading of the central corridor decreases, and the feasible routing set contracts accordingly. This forces the flow to redistribute over a broader subset of near-geodesic paths. The transition is mild at $\bar{\rho} = 0.6$, clearly visible at $\bar{\rho} = 0.3$, and strongest at $\bar{\rho} = 0.15$, where the central row alone can no longer sustain the full demand within the prescribed utilization limit.

In contrast to the K -sweep, the effect of varying $\bar{\rho}$ is not expressed primarily through a gradual change in objective curvature, but through a direct feasibility restriction. Consequently, the evolution of the profile is sharper and more threshold-like. Once the dominant central corridor approaches the admissibility boundary, rerouting is no longer merely favorable; it becomes necessary in order to maintain feasibility. This is precisely the mechanism that motivates the interpretation of $\bar{\rho}$ as a hard utilization constraint.

Since $K = 0$ throughout this sweep, the queueing-induced nonlinear terms vanish and the SO and UE objectives coincide. Accordingly, the two rows of Fig. 4 are effectively identical. This is useful from an interpretive standpoint, since it isolates the influence of $\bar{\rho}$ alone and shows that the observed broaden-

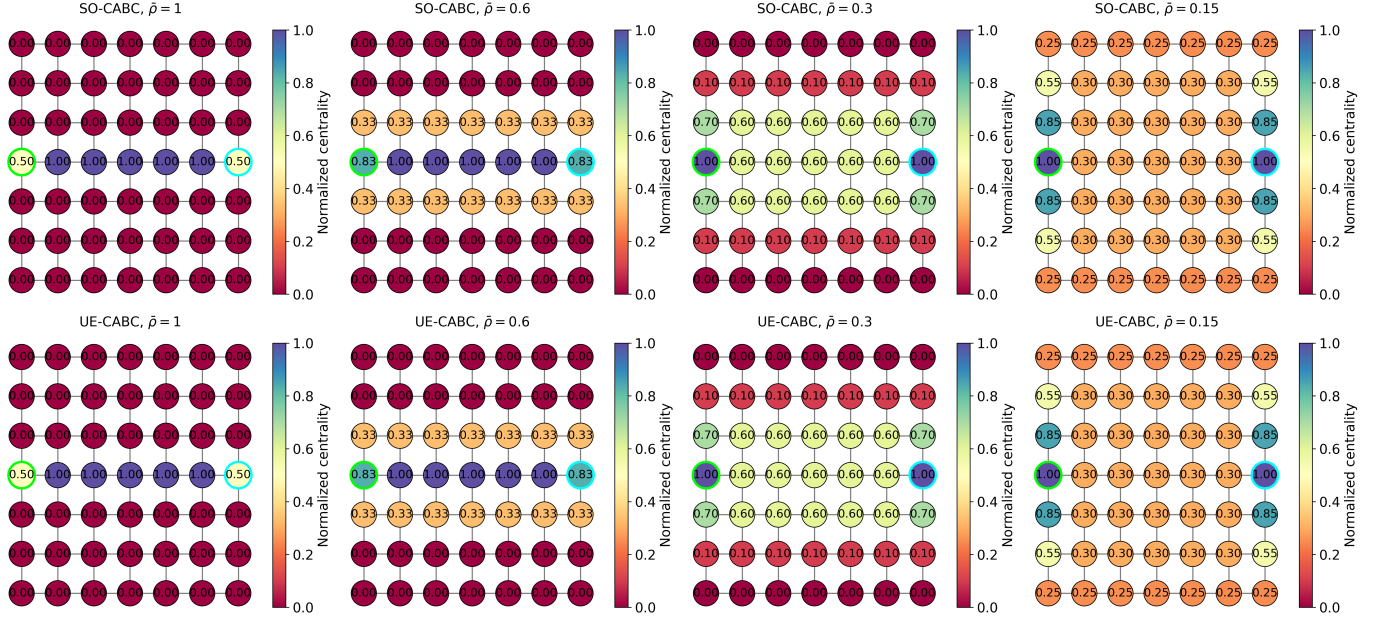
$\bar{\rho}$ -Sweep on a 7×7 Grid ($d = 100$, $K = 0$)

Fig. 4. Sensitivity of the induced centrality profiles to the utilization cap $\bar{\rho}$ on the 7×7 grid. The demand is fixed at $d = 100$ and the variability weight is fixed at $K = 0$. The top row shows SO-CABC and the bottom row shows UE-CABC. As $\bar{\rho}$ is tightened, the admissible loading of the central corridor decreases and traffic is forced to spread across a broader portion of the grid. Since $K = 0$, the SO and UE profiles coincide, isolating the hard feasibility-driven effect of the utilization constraint.

ing is driven entirely by the admissible-utilization constraint rather than by differences in the routing objective.

C. Discussion

The experiments of Figs. 3 and 4 reinforce the interpretation advanced in Section 6.2 and summarized in Remark 10 of the main paper. The two parameters K and $\bar{\rho}$ affect the induced centrality profiles in qualitatively different ways. Increasing K produces a gradual and distributed broadening of the profile, reflecting a soft increase in congestion sensitivity through the routing objective. By contrast, decreasing $\bar{\rho}$ produces sharper rerouting once the dominant corridor approaches the utilization boundary, reflecting the action of a hard feasibility constraint.

These experiments also indicate that the proposed centralities are qualitatively stable under moderate parameter variation. The dominant routing structure evolves in a predictable manner, and the induced profiles remain closely tied to the underlying physical interpretation of the model. In this sense, the sensitivity study supports the broader claim of the paper that CABC is not merely a fitted or heuristic ranking rule, but a dynamics-induced measure of node importance whose behavior remains interpretable as the congestion parameters are varied.

REFERENCES

- [1] B. T. Polyak, "Gradient methods for the minimisation of functionals," *USSR Computational Mathematics and Mathematical Physics*, vol. 3, no. 4, pp. 864–878, 1963.
- [2] H. Karimi, J. Nutini, and M. Schmidt, "Linear convergence of gradient and proximal-gradient methods under the polyak-lobasiewicz condition," in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 2016, pp. 795–811.



learning, learning-based

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